

Low-Complexity FPGA Implementation of Channel Estimation and Equalization for 100G Real-Time Upstream Burst-Mode Detection in Coherent PON

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Abstract We experimentally demonstrate low-complexity FPGA-based channel estimation and equalization for 100-Gb/s real-time upstream burst-mode detection in coherent PON. Reduced-parallelism one-tap SOP-aided MMSE decreases FPGA resource consumption by 35.8% versus full 128-parallel processing, with only 1-dB dynamic-range penalty and a 45.7-dB power budget. ©2026 The Author(s)

Introduction

The rapid growth of artificial intelligence services, mobile Internet, and high-definition video applications continues to drive the evolution of passive optical networks (PONs) [1]. As 25/50G NG-EPON and 50G-PON have been established, 100G and beyond have become the next important target for optical access [2]. Although intensity-modulation and direct-detection (IM/DD) solutions remain attractive because of low cost and architectural simplicity, scaling to higher single-wavelength capacities becomes increasingly challenging [3]. Coherent PON is therefore a promising candidate for future high-speed optical access because of its higher sensitivity and larger power budget [4].

However, coherent PON still faces critical challenges in upstream burst-mode reception [5]. Most reported studies focus on offline digital signal processing (DSP) [6], [7], while real-time field-programmable-gate-array (FPGA)-based implementation remains limited. In [8], real-time burst-mode digital coherent reception was first demonstrated for DSP-based upstream PON at 20 Gb/s using single-polarization QPSK burst signals. In [9], [10], a bidirectional FPGA-based real-time coherent TFDMA-PON was demonstrated with an upstream burst rate of 12.5 Gb/(s × subcarrier).

In our recent work [11], the world's first field trial of bidirectional 200G TFDM coherent PON enabled by real-time FPGA-based reception with two subcarriers was demonstrated. Despite these demonstrations, FPGA implementation details of burst-mode coherent DSP, particularly for channel estimation and equalization, remain limited in these works.

In this paper, we report the FPGA implementation of real-time channel estimation and equalization for a 100-Gb/s upstream burst-mode detection in coherent PON. Using a reduced-parallelism one-tap SOP-aided minimum-mean-square-error (MMSE) scheme, the channel estimation and equalization module significantly reduces FPGA resource consumption while maintaining the throughput required for real-time processing. Experimental results show that, compared with full 128-parallel processing, the 43-parallel implementation reduces LUT, FF, DSP, and BRAM consumption by 49.60%, 58.44%, 35.83%, and 64.91%, respectively, with only a 1-dB dynamic-range penalty, and achieves a maximum power budget of 45.7 dB.

Reduced-Parallelism FPGA Implementation of One-Tap SOP-Aided MMSE

Fig. 1(a) illustrates the FPGA-based upstream

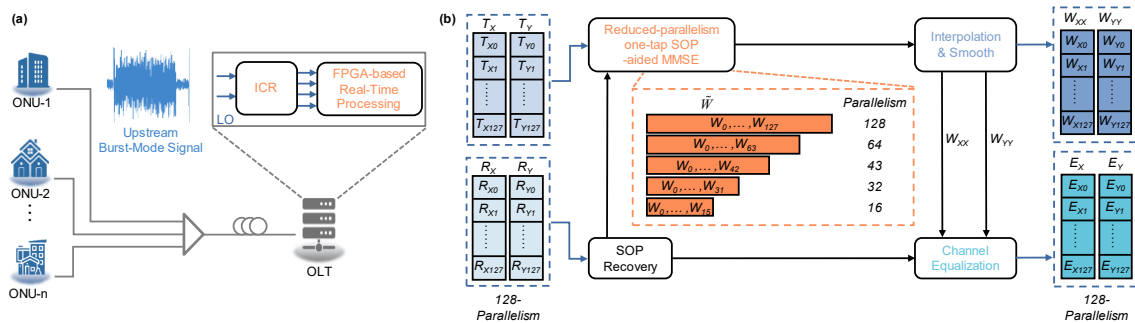


Fig. 1: (a) FPGA-based upstream real-time coherent PON architecture. (b) FPGA implementation of burst-mode channel estimation and equalization based on reduced-parallelism one-tap SOP-aided MMSE.

real-time coherent PON architecture, where up-stream burst-mode signals from multiple ONUs are received at the OLT and processed by FPGA-based real-time DSP after coherent detection. In such a burst-mode coherent receiver, channel estimation and equalization are key DSP functions that largely determine the system performance. Reference [12] gives the closed-form expressions of the training-sequence-based full 2×2 MMSE channel estimation coefficients. However, direct FPGA implementation of this full-MMSE algorithm is highly challenging because it requires joint calculation of four coefficients with cross-polarization terms under large parallelism. Therefore, as shown in Fig. 1(b), burst-mode channel estimation and equalization in this work is implemented using one-tap SOP-aided MMSE with reduced parallelism for real-time FPGA processing.

To reduce the implementation complexity, SOP recovery is first performed on the received data. After SOP recovery, the inter-polarization coupling is largely removed, and one-tap SOP-aided MMSE with reduced parallelism is applied in the coefficient-estimation path. Instead of using all 128-parallel recovered samples, only part of the recovered samples and the corresponding training symbols are selected for channel estimation. For an extraction interval N , the selected samples can be written as

$$\begin{aligned}\tilde{S}_X &= S_X(1:N:128), \tilde{S}_Y = S_Y(1:N:128) \\ \tilde{T}_X &= T_X(1:N:128), \tilde{T}_Y = T_Y(1:N:128)\end{aligned}\quad (1)$$

where $\tilde{S}_X, \tilde{S}_Y$ denote the extracted low-parallelism samples from the SOP-recovered data. $\tilde{T}_X$ and $\tilde{T}_Y$ the corresponding low-parallelism samples from the training sequences. These extracted samples correspond to 128/ N -parallel processing, e.g., 64, 43, 32, or 16. The corresponding low-parallelism coefficients are

calculated as

$$\tilde{W}_{XX} = \frac{E(\tilde{T}_X \tilde{S}_X^*)}{E(\tilde{S}_X \tilde{S}_X^*)}, \quad \tilde{W}_{YY} = \frac{E(\tilde{T}_Y \tilde{S}_Y^*)}{E(\tilde{S}_Y \tilde{S}_Y^*)} \quad (2)$$

and

$$\tilde{W}_{XY} = 0, \quad \tilde{W}_{YX} = 0 \quad (3)$$

The obtained low-parallelism coefficients are then interpolated back to the original 128-parallel structure, followed by 7-point smoothing to suppress coefficient fluctuation and improve equalization performance. In this way, the 128-parallel coefficients W_{XX} and W_{YY} are obtained from the low-parallelism estimates and used for channel equalization:

$$\begin{bmatrix} E_X \\ E_Y \end{bmatrix} = \begin{bmatrix} W_{XX} & 0 \\ 0 & W_{YY} \end{bmatrix} \begin{bmatrix} S_X \\ S_Y \end{bmatrix} \quad (4)$$

Accordingly, reduced parallelism is applied only to the coefficient-estimation path, while the equalization path remains 128-parallel, thereby significantly reducing the implementation complexity while maintaining the throughput required for real-time processing.

Real-Time Experimental Setup

Fig. 2 shows the experimental setup of the real-time burst-mode coherent PON system. At the transmitter side, 25-Gbaud QPSK burst-mode signals are generated by a 119.07072-GSa/s arbitrary waveform generator (AWG, Keysight M8194A) and modulated by a DP I/Q modulator driven by an external cavity laser (ECL) at 1552.524 nm. A variable optical attenuator (VOA) is used to emulate the optical splitting loss. At the receiver side, the burst-mode optical signal is detected by an integrated coherent receiver (ICR), and the received electrical signals are sampled by four ADC channels at 28.20096 GSa/s and processed by FPGA-based real-time DSP. The Rx DSP in Fig. 2(b) includes add overlap, I/Q de-

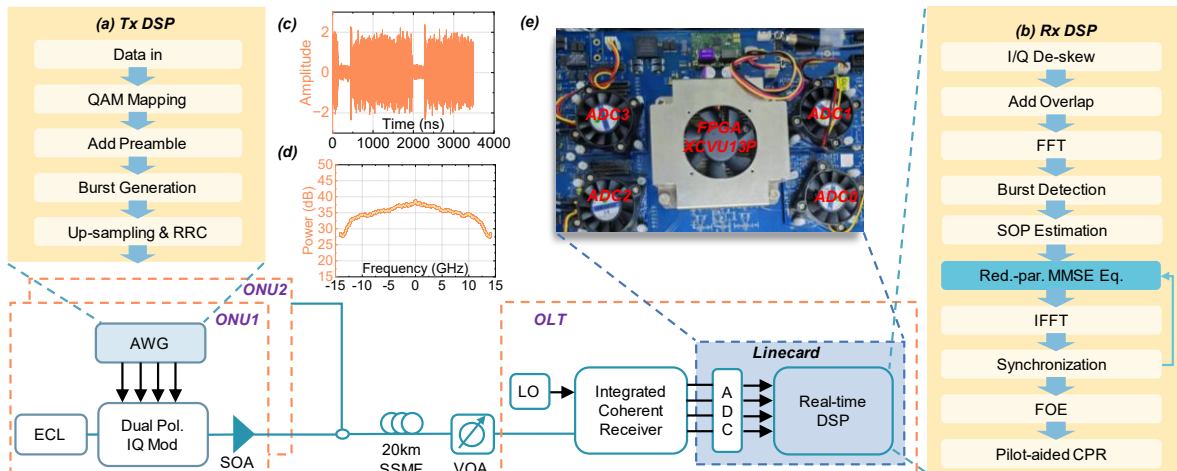


Fig. 2: Experimental setup of the real-time burst-mode coherent PON system. (a) shows the Tx DSP flow; (b) shows the FPGA-based real-time Rx DSP flow; (c) and (d) present the waveform and spectrum of burst-mode signal, respectively; (e) is the photograph of the real-time receiver hardware.

skew, FFT, burst detection, SOP estimation, reduced-parallelism channel estimation and equalization, IFFT, synchronization, frequency offset estimation (FOE), and pilot-aided carrier phase recovery (CPR). Fig. 2(c)–(e) show the burst-mode waveform, spectrum, and real-time receiver hardware, respectively.

Results and Discussions

Fig. 3 shows the BER versus received optical power (ROP) after 20-km transmission under different parallelism settings. When the parallelism is reduced from 128 to 43, only slight receiver-sensitivity degradation is observed. This is further summarized in Fig. 4, where the dynamic range decreases from 22 dB to 21 dB, corresponding to only 1 dB penalty, while the sensitivity degrades by only 0.6 dB.

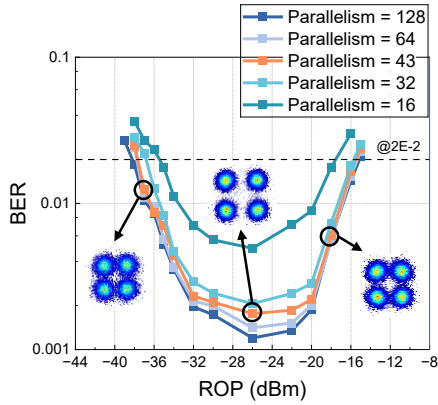


Fig. 3: BER versus received optical power (ROP) after 20-km transmission for different parallelism settings.

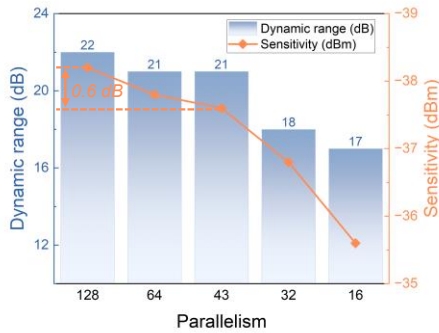


Fig. 4: Dynamic range and sensitivity under different parallelism settings

Fig. 5 shows the FPGA resource consumption under different parallelism settings. Compared with full 128-parallel processing, reducing the parallelism to 43 lowers the LUT, FF, DSP, and BRAM consumptions by 49.60%, 58.44%, 35.83%, and 64.91%, respectively, indicating a clear reduction in implementation complexity.

From the above resource and performance analyses, 43 parallelism provides a good trade-

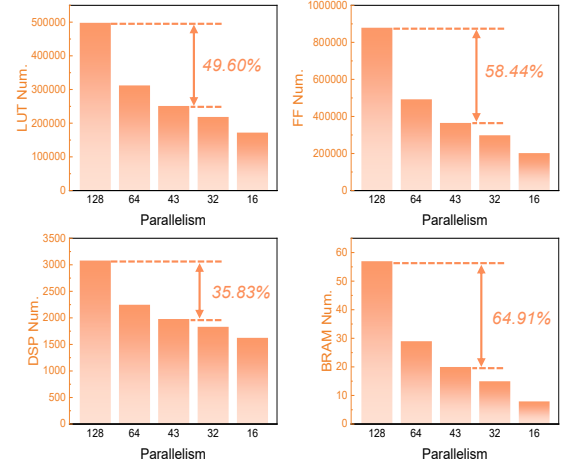


Fig. 5: FPGA resource consumption under different parallelism settings. (a), (b), (c), and (d) show the numbers of look-up tables (LUTs), flip-flops (FFs), digital signal processing (DSP) slices, and block RAMs (BRAMs), respectively.

off between hardware complexity and transmission performance. Based on this result, Fig. 6 shows the measured power budget under 43-parallel channel estimation and equalization. A maximum power budget of 45.7 dB is achieved at an SOA power of 10 dBm.

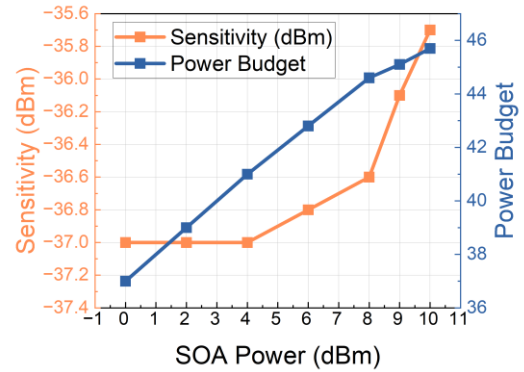


Fig. 6: Power budget of the real-time burst-mode coherent PON system with 43-parallel channel estimation and equalization.

Conclusions

In this paper, we demonstrate the real-time FPGA implementation of channel estimation and equalization for a 100-Gb/s burst-mode coherent PON based on reduced-parallelism one-tap SOP-aided MMSE. Real-time experiments show that 43 parallelism provides an effective trade-off between hardware complexity and performance, reducing LUT, FF, DSP, and BRAM consumption by 49.60%, 58.44%, 35.83%, and 64.91%, respectively, with only 1 dB dynamic-range penalty relative to full 128-parallel processing, while achieving a maximum power budget of 45.7 dB.

Acknowledgements

This work is supported by the Science and Technology Project of State Grid Corporation of China: Research and Application of Ultra-Low-Loss Ultra-Low-Latency Hollow-Core Fiber Cable for New Power System Communication Networks. (Project No.: 52180026001A-206-ZN).

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